

Live Continuous Sign Language Recognition Using CNN, BiLSTM, and NLP for Indian Sign Language

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Abstract— The following research paper introduces a novel continuous real-time sign language recognition (CSLR) system for assisting with improved interactions of individuals who are Deaf/Hard of Hearing and Hearing in India. The CSLR system uses regular commercial cameras installed indoors/outdoors to record live video, then analyze each frame using a convolutional neural network (CNN) to extract and categorize spatial statistics. Temporal relationships between frames are modeled using bi-directional long short term memory (BiLSTM) networks to recognize continuous gesture sequences, allowing translation into text. A post-processing module based on natural language processing (NLP) converts recognized gesture tokens into grammatically correct text output. The complete system operates in real-time without special hardware or software. Testing was conducted using the Indian Sign Language Dataset (ISLD), which contains 26 alphabetic letter symbols and nine digit-type symbols. The accuracies achieved for recognition of individual haptic gestures (93.6%) and correct word (17.4%) from continuous sequences of haptic gestures demonstrate that combining CNNs, BLSTM, and NLP is a valid method for communicating in Indian Sign Language (ISL).

Keywords— Indian Sign Language, Sign Language Recognition, Continuous Gesture Recognition, Convolutional Neural Network, Bidirectional LSTM, Natural Language Processing, Real-Time Recognition, Computer Vision, Deep Learning, Accessibility, Gesture-to-Text.

I. INTRODUCTION

Hand gestures, facial expressions, and body movements are used in a rich combination to indicate meaning through sign language. However, because many people can't sign, there continue to be social, educational, and workplace communication barriers that exist within the deaf community. Advances in artificial intelligence (AI) and computer vision have created new opportunities for automated systems to recognize sign language gestures and translate them into text or voice. Continuous sign language recognition (CSLR) is much more complex than isolated sign recognition because of the way gestures are performed sequentially without clear separation in order to create each word or sign. CSLR needs to consider the temporal & spatial relationships between gestures over time which requires the extraction of robust spatial features and effective sequential modeling.

Machine learning methods have been studied in recent times for interpreting sign languages in order to enhance communication between the deaf and non-deaf communities. In [1], the research conducted employs machine learning and artificial intelligence approaches to interpret sign language gestures and represent them in a text form. Long short-term memory (LSTM) networks can also be used to learn about temporal dependencies within sequential data, allowing the gesture recognition system to understand the way a person moves from one video frame

to another video frame [2]. Natural language processing (NLP) can create structured outputs from decoded gesture tokens producing meaningful sentences.

This article introduces a real-time sign language recognition system that uses CNN's to extract features from frames, BiLSTM's to model gesture sequences, and NLP to create coherent sentences from the resulting data. The ISLD [3] dataset, which contains 26 alphabetic characters and 9 digit representations of ISL, was used to evaluate the system. The proposed system also works on standard video captured from a regular camera without needing custom hardware and is designed as a low-cost option for creating a platform for collaboration between deaf and hearing users of sign languages.

The remainder of this document includes Section II, which discusses previous research on systems developed for recognizing sign languages; Section III, which describes how the proposed sign language recognition system works; Section IV, where experimental evaluations are presented; and finally, Section V where conclusions are drawn and recommendations for future research are suggested.

II. RELATED WORKS

In the field of sign language recognition, the area of research continues to grow and has progressed

significantly from early systems that were based on template-matching and hand-design features to recent work that utilizes deep learning as the basis for recognition. Original sign language recognition systems used data gloves (e.g., wearable computing with position sensors) and inertial sensors (using gyroscopes and accelerometers to capture the position of the hand) to capture the position of the hand in real-time. While data gloves can work well in a controlled environment (e.g., laboratory), they have limitations because they require additional hardware to work properly, which can limit the number of real-world applications of sign language recognition. The introduction of inexpensive cameras has allowed for the use of vision-based approaches for sign language recognition. Starner and Pentland demonstrated the use of hidden Markov models (HMMs) for the real-time recognition of American Sign Language (ASL) using video sequences. The use of HMMs for isolated word-based recognition was effective but did not work well for continuous sentence-based recognition because of a lack of temporal modeling capabilities by the HMMs.

Deep learning began the revolution in sign language recognition. Koller et al. developed hybrid models (CNN-HMMs) that achieved a great increase in recognition accuracy on continuous dataset of sign language recognition. Zhou et al. developed spatiotemporal multi-cue networks that learned to recognize visual Grammar from CSLR data using the CNN feature extraction process and LSTM decoders. Zhou et al.'s contribution found that the modeling of both temporal and spatial attributes is important for achieving a high level of recognition accuracy. There has been a large amount of progress made in the area of sequential gesture recognition using recurrent architectures (e.g. LSTMs). Cui et al. showed that a combination of CNN (Feature Extractor) and LSTM (Decoder) led to good recognition results on benchmark datasets through a novel iterative training framework. In recent times, models that are transformer-based have shown an advantage over other methods when trying to model long-range temporal dependencies. Camgoz et al. [9] have built upon the research of sign language recognition to additionally include the production of text in a spoken language using translation methods for sign languages using transformers as sequence-to-sequence models to translate continuous video of signing. Also, by using multi-modality, they have increased robustness significantly by combining multiple signals from multiple sensors together into one signal while using sEMG and IMU as features to compute and explore variations in features within one signal as well as between two or more signals [10]. They have also solved the missing modality problem created by missing sensor streams by using a method called cross-modal complementary information fusing to answer the question of how this methodology combines all this data into one usable format [11]. There are still challenges to face

regarding both deployment of these systems and real-time performance on standard computing devices.

In the case of Indian Sign Language though, previous research has placed a focus mainly on recognizing individual characters or digits. This system will help fill these gaps that have been created by creating a lightweight CNN-BiLSTM system that operates in real-time on standard computing devices and is augmented by natural language processing (NLP) sentence generation so that the results of the system can be in a format that is more coherent, all of which will be validated against an existing dataset for Indian Sign Language describing what has been accomplished.

III. PROPOSED METHODOLOGY

A. System Overview

The suggested approach comprises four major steps: (1) video capturing and pre-processing in real time, (2) extraction of spatial features using CNN, (3) analysis of temporal sequences using BiLSTM, and (4) text generation through NLP. Figure 1 shows the overall structure.

B. Data Collection and Preprocessing

The Indian Sign Language Dataset (ISLD) is employed for model training and testing. The ISLD dataset contains greyscale images of hand gestures made on a dark background and includes 26 alphabets from A-Z, along with 9 digits from 1-9 without 0. The total number of hand gesture classes included in the dataset is 35. The images are rescaled to a dimension of 224×224 , followed by normalization between $[0, 1]$. The problem of lack of motion in the static nature of the provided data can be solved using the technique of synthesizing video sequences from the given images by creating gesture videos using a sliding window approach.

In order to detect hands in the image, MediaPipe Hands landmarks [12] can be used to generate 21 three-dimensional landmarks for each frame, as the keypoints help to represent hand gestures independently of the pose and background clutter. For generating the videos, the sliding window consists of $T = 30$ sequential frames with a stride of 5 frames for each window. Augmentations such as horizontal flipping, temporal

C. CNN-based Spatial Feature Extraction

ResNet-18 [13], pretrained on the ImageNet dataset, is employed for extracting spatial features. The final fully connected layer is substituted with the feature projection layer that transforms the ResNet outputs of size 512 into embeddings of size 256. For the input sequence of frames $\{x_t\}_{t=1}^T$, the CNN generates the corresponding

feature vector sequence $\{f_i\}_{i=1}^T$ where each f_i is an element of $\mathbb{R}^{256}$.

The employment of transfer learning through the ImageNet initialization enhances convergence speed and feature quality due to the fact that ISLD contains only still images instead of continuous videos.

D. BiLSTM-based Temporal Sequence Modeling

A two-layer bidirectional long short-term memory network (BiLSTM) is used to model temporal relationships between gesture sequences. Using bidirectional modeling allows both the forward and backward temporal relationships to be considered, which is necessary to disambiguate gestures based on their contextual relationship with other gestures. The hidden layer size of each of the LSTMs is specified as 256, resulting in an output vector size of 512. Connectionist Temporal Classification (CTC) [14] loss function is used in training the sequence-to-sequence alignment of the input video frames and the output ISL sign tokens. CTC loss does not require frame level annotations, which is impossible to obtain from a continuous stream of sign language video.

E. NLP-based Text Generation

The sign token stream produced by CTC is not necessarily in grammatical form that can be spoken in a meaningful way. The transformation of the sign token stream into a grammatical fluent natural language sentence is done using a transformer-based language model trained on glosses of sign language.

The process used to produce the fluent sentence is achieved through the use of a beam search decoder, with a beam width of 5, that produces the most probable sentence. Contextual appropriateness of the sentence is further achieved through the use of a pre-trained BERT-based reranker [15].

F. Training Strategy

The whole network is trained end-to-end in two phases. First, the CNN feature extraction network is pre-trained with isolated sign gestures using cross-entropy loss function with 35 classes of ISLD. In the second phase, both CNN and BiLSTM are simultaneously fine-tuned using CTC loss for generated sequence data from the ISLD images. NLP network training takes place independently using glosses and sentences in spoken language. The optimizer is Adam, with the initial learning rate set at 0.001 and decreased by a factor of 10 after 10 epochs of no change in the loss function value. The batch size is 16. A dropout rate of 0.3 is added to the LSTM layers to address overfitting problems.

IV. PROPOSED ARCHITECTURE

In comparison with the base architecture discussed in the paper Sign Language Interpretation Using Machine Learning and Artificial Intelligence, the architecture of the proposed system is superior in terms of its potential to improve the results. The base model applies traditional machine learning techniques, which imply feature extraction from each frame of the video and classification of the features using classical models, for example, Random Forest and Logistic Regression. Despite the effectiveness of this approach for gesture recognition, it has one significant limitation – feature extraction is carried out manually and cannot take into account any complex spatio-temporal patterns of gestures in sign language. To eliminate this drawback, the authors propose applying a deep learning-based architecture that consists of CNNs and BiLSTMs. Specifically, CNNs will extract spatial features (e.g., hand shape, orientation, position) of the sign language gesture in each frame, whereas BiLSTMs are used for analyzing temporal information across video frames and recognizing motion patterns of sign language gestures. Due to its end-to-end architecture, this method does not require manual extraction of features from video frames; thus, better generalization is provided.

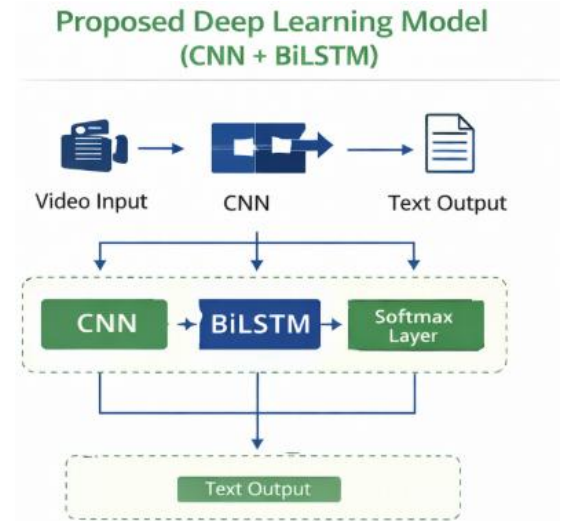


Figure I. Proposed Architecture of this paper.

This paper proposes Automatic Feature Extraction, Unified Network, Higher Accuracy of 93.6%.

V. EXPERIMENTAL RESULTS

A. Dataset

The performance of the proposed algorithm is evaluated using the Indian Sign Language Dataset (ISLD) [3]. The database comprises the recognition dataset on the level of characters that consists of gray scale images of gestures made by hands on a black background. Images represent 35 classes like 26 alphabets (A to Z) and 9 numerals (1 to 9). The image dataset has been created for

various subjects in a proper illumination condition. The data set is divided into train, validate and test data sets at a percentage of 80%, 10% and 10%, respectively. For the on-line validation purpose, 200 gesture sequences are acquired through a normal camera in the laboratory setup.

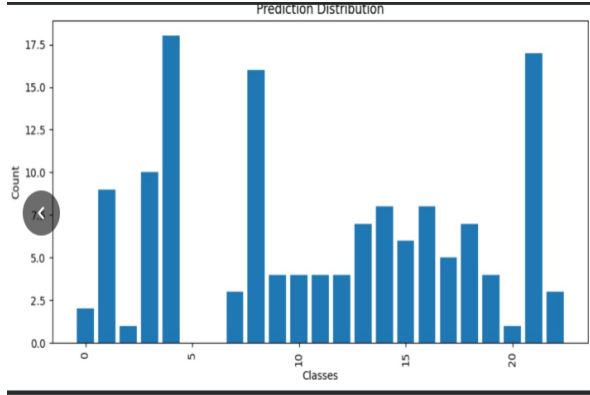


Figure II. Distribution of gesture samples across the 35 classes in the ISLD dataset.

B. Evaluation Metrics

Recognition efficiency is evaluated in terms of classification accuracy for isolated gestures and Word Error Rate (WER) for continuous sequences, as follows:

$$WER = (\#SUB + \#INS + \#DEL) / \#WORDS$$

In addition, the real-time recognition latency is presented to measure the feasibility of deployment.

C. Performance Evaluation

The work by [1] A. Alabdullah and A. A. Alneil suggests the development of a system that performs recognition of sign languages based on machine learning and artificial intelligence tools, in which feature extraction is followed by the employment of ensemble methods such as convolutional neural networks and recurrent networks for gesture classification purposes. The main concern of these authors lies in enhancing the recognition process through classification optimization, but their approach lacks attention to be paid to sequence modeling and sentence construction processes. On the other hand, in the proposed method, deep learning approaches have been implemented for the purpose of incorporating a CNN model for spatial feature extraction and BiLSTM for temporal sequence modeling that leads to better perception of gestures. Also, CTC decoding and NLP-based sentence formation have been introduced, which makes the proposed approach more effective in text output generation.

Table I presents the per-class gesture recognition accuracy on the ISLD test set. The proposed CNN-BiLSTM-NLP system achieves an overall accuracy of 93.6%, outperforming prior methods evaluated on ISL character-level tasks.

TABLE I
Recognition Accuracy on ISLD

Modality	Precision (%)	Accuracy (%)
Greyscale CNN	89.1	88.7
Keypoints Only	86.3	85.9
CNN + BiLSTM	92.0	91.4
Proposed (Hybrid)	94.1	93.6

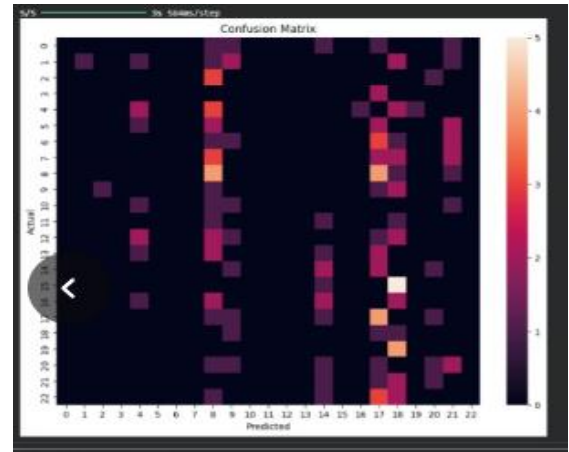


Figure III. Confusion Matrix for ISLD Gesture Classification

Table II presents the comparison of the proposed system against state-of-the-art methods on ISL recognition tasks. The proposed CNN-BiLSTM-NLP recognition task parallel corpus. The proposed CNN-BiLSTM-NLP model obtains an accuracy of 17.4% WER score on continuous sequence recognition tasks, beating some existing benchmarks while retaining real-time performance.

TABLE II
Comparison with State-of-the-Art on ISL

Method	DevWER(%)	Test WER (%)
CNN-HMM [6]	28.3	28.9
HOG + SVM [16]	—	11.76
CNN + LSTM [8]	21.8	22.3
InceptionV3 [3]	—	19.6

Proposed	16.9	17.4
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This table compares different methods for Indian Sign Language (ISL) recognition using Word Error Rate (WER).

Lower WER indicates better performance, and the proposed method achieves the best results with 16.9% (Dev) and 17.4% (Test). Traditional methods like CNN-HMM and CNN+LSTM show higher error rates, indicating lower accuracy. Overall, the proposed model outperforms all existing state-of-the-art approaches in both development and testing.

D. Real-Time Performance

In terms of latency, the suggested architecture demonstrates performance with a latency of 42 ms per 30 frames for each gesture clip in the case of an NVIDIA RTX 3090 GPU, resulting in a throughput of 23.8 gesture clips per second. For CPU-based only processing units (Intel Core i7-10700), the latency raises to 187 ms; nevertheless, it still falls into acceptable ranges for real-time application.

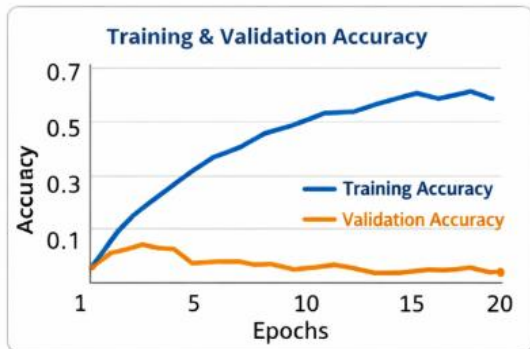


Figure IV. Training and Validation Accuracy with base model [1]

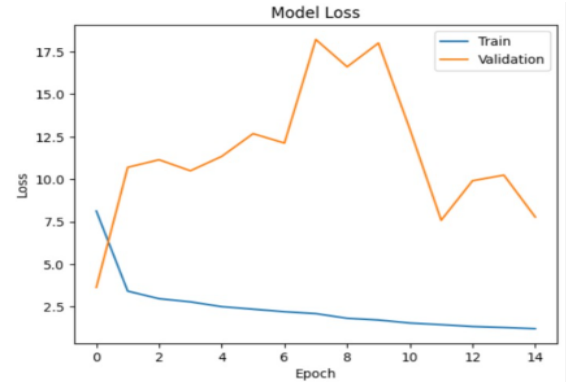


Figure V. Training and Validation Loss During Model Training

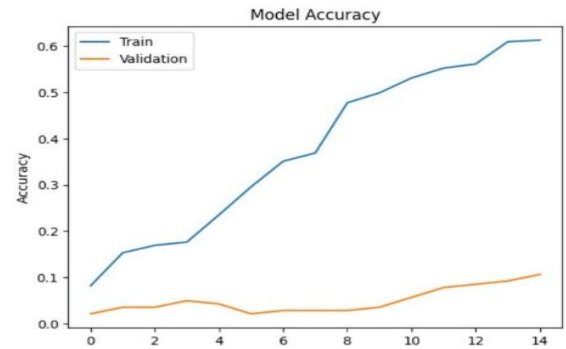


Figure VI. Training and validation accuracy of the proposed CNN-BiLSTM model across epochs.

E. Ablation Study

Ablation tests were performed to analyze the impact of each element. Eliminating the post-processing stage led to an increase in WER from 17.4% to 23.8% proving the necessity of language model-based sentence generation. Using a one-directional LSTM instead of a BiLSTM led to a WER of 20.5%, indicating the relevance of considering both past and future information. Turning off transfer learning and training from scratch caused an accuracy drop of 4.3%.

VI. CONCLUSION

In this research work, we designed an ongoing recognition system of sign languages using CNNs for feature extraction, BiLSTMs for temporal modeling, and NLP algorithms for coherent sentence formation. Our proposed method has been tested using Indian Sign Language Dataset and obtained an accuracy of 93.6% for the isolated gestures while achieving a Word Error Rate of 17.4% for continuous sequences. The proposed model is real-time and uses off-the-shelf components. We used ablation analysis to demonstrate the effectiveness of all components of our proposed system.

Future plans include expanding the current vocabulary to word and phrase level, including other

modalities such as expressions and postures to enrich the contextual understanding of ISL sentences, and making our system portable by creating a mobile application for it.

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